Appendix D4 (from Petrucci textbook)

TABLE D.4 Standard Electrode (Reduction) Potentials at 25 °C[a]

Reduction Half-Reaction	Cell Notation	Standard Reduction Potential, V
$F_2(g) + 2\,e^- \rightleftharpoons 2\,F^-(aq)$	$F^- \mid F_2 \mid Pt$	+2.866
$OF_2(g) + 2\,H^+(aq) + 4\,e^- \rightleftharpoons H_2O(l) + 2\,F^-(aq)$	$F^-, H^+ \mid OF_2 \mid Pt$	+2.1
$O_3(g) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons O_2(g) + H_2O(l)$	$H^+, H_2O \mid O_3, O_2 \mid Pt$	+2.075
$S_2O_8^{2-}(aq) + 2\,e^- \rightleftharpoons 2\,SO_4^{2-}(aq)$	$SO_4^{2-}, S_2O_8^{2-} \mid Pt$	+2.01
$Ag^{2+}(aq) + 2e^- \rightleftharpoons Ag^+(aq)$	$Ag^+, Ag^{2+} \mid Pt$	+1.98
$Co^{3+}(aq) + e^- \rightleftharpoons Co^{2+}(aq)$	$Co^{2+}, Co^{3+} \mid Pt$	+1.92
$H_2O_2(aq) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons 2\,H_2O(l)$	$H_2O_2, H^+ \mid Pt$	+1.763
$Ce^{4+}(aq) + e^- \rightleftharpoons Ce^{3+}(aq)$	$Ce^{3+}, Ce^{4+} \mid Pt$	+1.72
$MnO_4^-(aq) + 4\,H^+(aq) + 3\,e^- \rightleftharpoons MnO_2(s) + 2\,H_2O(l)$	$MnO_2 \mid MnO_4^-, H^+ \mid Pt$	+1.70
$PbO_2(s) + SO_4^{2-}(aq) + 4\,H^+(aq) + 2\,e^- \rightleftharpoons PbSO_4(s) + 2\,H_2O(l)$	$PbSO_4, PbO_2 \mid SO_4^{2-}, H^+ \mid Pt$	+1.69
$Mn^{3+}(aq) + e^- \rightleftharpoons Mn^{2+}(aq)$	$Mn^{2+}, Mn^{3+} \mid Pt$	+1.54
$Au^{3+}(aq) + 3\,e^- \rightleftharpoons Au(s)$	$Au^{3+} \mid Au$	+1.52
$MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \rightleftharpoons Mn^{2+}(aq) + 4\,H_2O(l)$	$Mn^{2+}, MnO_4^- \mid Pt$	+1.51
$2\,BrO_3^-(aq) + 12\,H^+(aq) + 10\,e^- \rightleftharpoons Br_2(l) + 6\,H_2O(l)$	$BrO_3^-, H^+ \mid Br_2 \mid Pt$	+1.478
$PbO_2(s) + 4\,H^+(aq) + 2\,e^- \rightleftharpoons Pb^{2+}(aq) + 2\,H_2O(l)$	$Pb^{2+}, H^+ \mid PbO_2 \mid Pt$	+1.455
$ClO_3^-(aq) + 6\,H^+(aq) + 6\,e^- \rightleftharpoons Cl^-(aq) + 2\,H_2O(l)$	$Cl^-, ClO_3^-, H^+ \mid Pt$	+1.450
$Au^{3+}(aq) + 2\,e^- \rightleftharpoons Au^+(aq)$	$Au^+, Au^{3+} \mid Pt$	+1.36

(continued)

Reduction Half-Reaction	Cell Notation	Standard Reduction Potential, V
$Cl_2(g) + 2e^- \rightleftharpoons 2\,Cl^-(aq)$	$Cl^- \mid Cl_2 \mid Pt$	+1.36
$Cr_2O_7^{2-}(aq) + 14\,H^+(aq) + 16\,e^- \rightleftharpoons 2\,Cr^{3+}(aq) + 7\,H_2O(l)$	$Cr^{3+}, Cr_2O_7^{2-} \mid Pt$	+1.32
$Tl^{3+}(aq) + 2e^- \rightleftharpoons Tl^+(aq)$	$Tl^+, Tl^{3+} \mid Pt$	+1.252
$MnO_2(s) + 4\,H^+(aq) + 2\,e^- \rightleftharpoons Mn^{2+}(aq) + 2\,H_2O(l)$	$Mn^{2+}, H^+ \mid MnO_2 \mid Pt$	+1.23
$O_2(g) + 4\,H^+(aq) + 4\,e^- \rightleftharpoons 2\,H_2O(l)$	$H_2O \mid O_2 \mid Pt$	+1.229
$IO_3^-(aq) + 12\,H^+(aq) + 10\,e^- \rightleftharpoons I_2(s) + 6\,H_2O(l)$	$IO_3^-, H^+ \mid I_2 \mid Pt$	+1.20
$ClO_4^-(aq) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons ClO_3^-(aq) + H_2O(l)$	$ClO_4^-, H^+, ClO_3^- \mid Pt$	+1.189
$ClO_3^-(aq) + 2\,H^+(aq) + e^- \rightleftharpoons ClO_2(g) + H_2O(l)$	$ClO_3, H^+ \mid ClO_2^- \mid Pt$	+1.175
$NO_2(g) + H^+(aq) + e^- \rightleftharpoons HNO_2(aq)$	$HNO_2, H^+ \mid NO \mid Pt$	+1.07
$Br_2(l) + 2\,e^- \rightleftharpoons 2\,Br^-$	$Br^-, Br_2 \mid Pt$	+1.065
$NO_2(g) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons NO(g) + H_2O(l)$	$NO_2, NO \mid Pt$	+1.03
$AuCl_4^-(aq) + 3\,e^- \rightleftharpoons Au(s) + 4\,Cl^-(aq)$	$AuCl_4^-, Cl^- \mid Au$	+1.002
$VO_2^+(aq) + 2\,H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l)$	$VO^{2+}, VO_2^+, H^+ \mid Pt$	+1.000
$NO_3^-(aq) + 4\,H^+(aq) + 3\,e^- \rightleftharpoons NO(g) + 2\,H_2O(l)$	$NO_3^-, H^+ \mid NO \mid Pt$	+0.956
$Cu^{2+}(aq) + I^-(aq) + e^- \rightleftharpoons CuI(s)$	$Cu^{2+}, I^- \mid CuI \mid Pt$	+0.86
$2\,Hg^{2+}(aq) + 2\,e^- \rightleftharpoons Hg_2^{2+}(aq)$	$Hg_2^{2+}, Hg^{2+} \mid Pt$	+0.92
$2\,Hg^{2+}(aq) + 2\,e^- \rightleftharpoons Hg(l)$	$Hg^{2+} \mid Hg$	+0.854
$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	$Ag^+ \mid Ag$	+0.800
$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	$Fe^{2+}, Fe^{3+} \mid Pt$	+0.771
$O_2(g) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2O_2(aq)$	$H_2O_2, H^+ \mid O_2 \mid Pt$	+0.695
$2\,HgCl_2(aq) + 2\,e^- \rightleftharpoons Hg_2Cl_2(s) + Cl^-(aq)$	$Cl^-, HgCl_2 \mid Hg_2Cl_2 \mid Pt$	+0.63
$MnO_4^-(aq) + e^- \rightleftharpoons MnO_4^{2-}(aq)$	$MnO_4^{2-}, MnO_4^- \mid Pt$	+0.56
$I_2(s) + 2\,e^- \rightleftharpoons 2\,I^-(aq)$	$I^- \mid I_2 \mid Pt$	+0.535
$Cu^+(aq) + e^- \rightleftharpoons Cu(s)$	$Cu^+ \mid Cu$	+0.520
$H_2SO_3(aq) + 4\,H^+(aq) + 4\,e^- \rightleftharpoons S(s) + 2\,H_2O(l)$	$H_2SO_3, H^+ \mid S \mid Pt$	+0.449
$O_2(g) + 2\,H_2O(1) + 4e^- \rightleftharpoons 4\,OH^-$	$OH^- \mid O_2 \mid Pt$	+0.401
$C_2N_2(g) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons 2\,HCN(aq)$	$HCN, H^+ \mid C_2N_2 \mid Pt$	+0.37
$Fe(CN)_6^{3-}(aq) + e^- \rightleftharpoons Fe(CN)_6^{4-}(aq)$	$Fe(CN)_6^{4-}, Fe(CN)_6^{3-} \mid Pt$	+0.361
$Cu^{2+}(aq) + 2\,e^- \rightleftharpoons Cu(s)$	$Cu^{2+} \mid Cu$	+0.340
$VO^{2+}(aq) + 2\,H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l)$	$V^{3+}, VO^{2+}, H^+ \mid Pt$	+0.337
$PbO_2(s) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons PbO(s) + H_2O(l)$	$H^+ \mid PbO_2, PbO \mid Pt$	+0.28
$Hg_2Cl_2(s) + 2\,e^- \rightleftharpoons 2\,Hg(l) + 2\,Cl^-(aq)$	$Cl^- \mid Hg_2Cl_2 \mid Hg \mid Pt$	+0.2676
$HAsO_2(aq) + 3\,H^+(aq) + 3\,e^- \rightleftharpoons As(s) + 2\,H_2O(l)$	$HAsO_2, H^+ \mid As$	+0.240
$AgCl(s) + e^- \rightleftharpoons Ag(s) + Cl^-$	$Cl^- \mid AgCl \mid Ag$	+0.2223
$SO_4^{2-}(aq) + 4\,H^+(aq) + 2\,e^- \rightleftharpoons 2\,H_2O(l) + SO_2(g)$	$SO_4^{2-}, H^+ \mid SO_2 \mid Pt$	+0.17
$Cu^{2+}(aq) + e^- \rightleftharpoons Cu^+(aq)$	$Cu^+, Cu^{2+} \mid Pt$	+0.159
$Sn^{4+}(aq) + 2\,e^- \rightleftharpoons Sn^{2+}(aq)$	$Sn^{2+}, Sn^{4+} \mid Pt$	+0.154
$S(s) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2S(g)$	$H^+ \mid H_2S \mid S$	+0.144
$AgBr(s) + e^- \rightleftharpoons Ag(s) + Br^-(aq)$	$Br^- \mid AgBr \mid Ag$	+0.071
$2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2(g)$	$H^+ \mid H_2 \mid Pt$	0.0
$Pb^{2+}(aq) + 2\,e^- \rightleftharpoons Pb(s)$	$Pb^{2+} \mid Pb$	−0.125
$Sn^{2+}(aq) + 2\,e^- \rightleftharpoons Sn(s)$	$Sn^{2-} \mid Sn$	−0.137
$AgI(s) + e^- \rightleftharpoons Ag(s) + I^-(aq)$	$I^+ \mid AgI \mid Ag$	−0.152
$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$	$V^{3+}, V^{2+} \mid Pt$	−0.255
$Ni^{2+}(aq) + 2\,e^- \rightleftharpoons Ni(s)$	$Ni^{2+} \mid Ni$	−0.257
$H_3PO_4(aq) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons H_3PO_3(aq) + H_2O(l)$	$H_3PO_3, H_3PO_4, H^+ \mid Pt$	−0.276
$Co^{2+}(aq) + 2\,e^- \rightleftharpoons Co(s)$	$Co^{2+} \mid Co$	−0.277
$Tl^+ + e^- \rightleftharpoons Tl$	$Tl^+ \mid Tl$	−0.336
$In^{3+}(aq) + 3\,e^- \rightleftharpoons In(s)$	$In^{3+} \mid In$	−0.338

Reduction Half-Reaction	Cell Notation	Standard Reduction Potential, V
$PbSO_4(s) + 2\,e^- \rightleftharpoons Pb(s) + SO_4^{2-}$	$SO_4^{2-} \mid PbSO_4 \mid Pb$	−0.356
$Cd^{2+}(aq) + 2\,e^- \rightleftharpoons Cd$	$Cd^{2+} \mid Cd$	−0.403
$Cr^{3+}(aq) + e^- \rightleftharpoons Cr^{2+}(aq)$	$Cr^{2+}, Cr^{3+} \mid Pt$	−0.424
$Fe^{2+}(aq) + 2\,e^- \rightleftharpoons Fe(s)$	$Fe^{2+} \mid Fe$	−0.440
$2\,CO_2(aq) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2C_2O_4(aq)$	$H_2C_2O_4, CO_2, H^+ \mid Pt$	−0.49
$Zn^{2+}(aq) + 2\,e^- \rightleftharpoons Zn(s)$	$Zn^{2+} \mid Zn$	−0.763
$2\,H_2O(l) + 2\,e^- \rightleftharpoons H_2(g) + 2\,OH^-(aq)$	$OH^- \mid H_2 \mid Pt$	−0.828
$Cr^{2+}(aq) + 2\,e^- \rightleftharpoons Cr(s)$	$Cr^{2+} \mid Cr$	−0.90
$Mn^{2+}(aq) + 2\,e^- \rightleftharpoons Mn(s)$	$Mn^{2+} \mid Mn$	−1.180
$Al^{3+}(aq) + 3\,e^- \rightleftharpoons Al(s)$	$Al^{3+} \mid Al$	−1.662
$Mg^{2+}(aq) + 2\,e^- \rightleftharpoons Mg(s)$	$Mg^{2+} \mid Mg$	−2.372
$Na^+(aq) + e^- \rightleftharpoons Na(s)$	$Na^+ \mid Na$	−2.714
$Ca^{2+}(aq) + 2\,e^- \rightleftharpoons Ca(s)$	$Ca^{2+} \mid Ca$	−2.868
$Sr^{2+}(aq) + 2\,e^- \rightleftharpoons Sr(s)$	$Sr^{2+} \mid Sr$	−2.899
$Ba^{2+}(aq) + 2\,e^- \rightleftharpoons Ba(s)$	$Ba^{2+} \mid Ba$	−2.905
$K^+(aq) + e^- \rightleftharpoons K(s)$	$K^+ \mid K$	−2.931
$Li^+(aq) + e^- \rightleftharpoons Li(s)$	$Li^+ \mid Li$	−3.05
Basic Solution		
$O_3(g) + H_2O(l) + 2\,e^- \rightleftharpoons O_2(g) + 2\,OH^-(aq)$	$OH^- \mid O_3, O_2 \mid Pt$	+1.246
$ClO^-(aq) + H_2O(l) + 2\,e^- \rightleftharpoons Cl^-(aq) + 2\,OH^-(aq)$	$OH^-, ClO_3^-, Cl^- \mid Pt$	+0.890
$H_2O_2(aq) + 2\,e^- \rightleftharpoons 2\,OH^-(aq)$	$OH^-, H_2O_2 \mid Pt$	+0.88
$BrO^-(aq) + H_2O(l) + 2\,e^- \rightleftharpoons Br^-(aq) + 2\,OH^-(aq)$	$OH^-, BrO^-, Br^- \mid Pt$	+0.766
$ClO_3^-(aq) + 3\,H_2O(l) + 6\,e^- \rightleftharpoons Cl^-(aq) + 6\,OH^-(aq)$	$OH^-, ClO_3^-, Cl^- \mid Pt$	+0.622
$2\,AgO(s) + H_2O(l) + 2\,e^- \rightleftharpoons Ag_2O(s) + 2\,OH^-(aq)$	$OH^- \mid Ag_2O \mid AgO \mid Pt$	+0.604
$MnO_4^-(aq) + 2\,H_2O(l) + 3\,e^- \rightleftharpoons MnO_2(s) + 4\,OH^-(aq)$	$OH^-, MnO_4^- \mid MnO_2 \mid Pt$	+0.60
$BrO_3^-(aq) + 3\,H_2O(l) + 6\,e^- \rightleftharpoons Br^-(aq) + 6\,OH^-(aq)$	$OH^-, BrO_3^-, Br^- \mid Pt$	+0.584
$2\,BrO^-(aq) + 2\,H_2O(l) + 2\,e^- \rightleftharpoons Br_2(l) + 4\,OH^-(aq)$	$OH^-, BrO^- \mid Br_2 \mid Pt$	+0.455
$2\,IO^-(aq) + 2\,H_2O(l) + 2\,e^- \rightleftharpoons I_2(s) + 4\,OH^-(aq)$	$OH^-, IO^- \mid I_2 \mid Pt$	+0.42
$O_2(g) + 2\,H_2O(l) + 4\,e^- \rightleftharpoons 4\,OH^-(aq)$	$OH^- \mid O_2 \mid Pt$	+0.401
$Ag_2O(s) + H_2O(l) + 2\,e^- \rightleftharpoons 2\,Ag(s) + 2\,OH^-(aq)$	$OH^-, Ag_2O \mid Ag$	+0.342
$Co(OH)_3(s) + e^- \rightleftharpoons Co(OH)_2(s) + OH^-(aq)$	$OH^- \mid Co(OH)_3 \mid Co(OH)_2 \mid Pt$	+0.17
$2\,MnO_2(s) + H_2O(l) + 2\,e^- \rightleftharpoons Mn_2O_3(s) + 2\,OH^-(aq)$	$OH^-, MnO_2 \mid Mn_2O_3 \mid Pt$	+0.118
$NO_3^-(aq) + H_2O(l) + 2\,e^- \rightleftharpoons NO_2^-(aq) + 2\,OH^-(aq)$	$OH^-, NO_2^-, NO_3^- \mid Pt$	+0.01
$CrO_4^{2-}(aq) + 4\,H_2O(l) + 3\,e^- \rightleftharpoons Cr(OH)_3(s) + 5\,OH^-(aq)$	$OH^-, CrO_4^{2-} \mid Cr(OH)_3 \mid Pt$	−0.11
$S(s) + 2\,e^- \rightleftharpoons S^{2-}(aq)$	$S^{2-} \mid S \mid Pt$	−0.48
$HPbO_2^-(aq) + H_2O(l) + 2\,e^- \rightleftharpoons Pb(s) + 3\,OH^-(aq)$	$OH^-, HPbO_2^- \mid Pb$	−0.54
$HCHO(aq) + 2\,H_2O(l) + 2\,e^- \rightleftharpoons CH_3OH(aq) + 2\,OH^-(aq)$	$OH^-, HCHO, CH_3OH \mid Pt$	−0.59
$SO_3^{2-}(aq) + 3\,H_2O(l) + 4\,e^- \rightleftharpoons S(s) + 6\,OH^-(aq)$	$OH^-, SO_3^{2-} \mid S \mid Pt$	−0.66
$AsO_4^{3-}(aq) + 2\,H_2O(l) + 2\,e^- \rightleftharpoons AsO_2^-(aq) + 4\,OH^-(aq)$	$OH^-, AsO_4^{3-}, AsO_2^- \mid Pt$	−0.67
$AsO_2^-(aq) + 2\,H_2O(l) + 3\,e^- \rightleftharpoons As(s) + 4\,OH^-(aq)$	$OH^-, SO_2^- \mid As \mid Pt$	−0.68
$Cd(OH)_2(s) + 2\,e^- \rightleftharpoons Cd(s) + 2\,OH^-(aq)$	$OH^- \mid Cd(OH)_2 \mid Cd \mid Pt$	−0.824
$2\,H_2O(l) + 2\,e^- \rightleftharpoons H_2(g) + 2\,OH^-(aq)$	$OH^- \mid H_2 \mid Pt$	−0.828
$OCN^-(aq) + H_2O(l) + 2\,e^- \rightleftharpoons CN^-(aq) + 2\,OH^-(aq)$	$OH^-, CN^-, OCN^- \mid Pt$	−0.97
$As(s) + 3\,H_2O(l) + 3\,e^- \rightleftharpoons AsH_3(g) + 3\,OH^-(aq)$	$OH^- \mid As \mid AsH_3 \mid Pt$	−1.21
$Zn(OH)_2(s) + 2\,e^- \rightleftharpoons Zn(s) + 2\,OH^-(aq)$	$OH^- \mid Zn(OH)_2 \mid Zn$	−1.246
$Sb(s) + 3\,H_2O(l) + 3\,e^- \rightleftharpoons SbH_3(g) + 3\,OH^-(aq)$	$OH^- \mid Sb \mid SbH_3 \mid Pt$	−1.338
$Al(OH)_4^-(aq) + 3\,e^- \rightleftharpoons Al(s) + 4\,OH^-(aq)$	$OH^-, Al(OH)_4^- \mid Al$	−2.310
$Mg(OH)_2(s) + 2\,e^- \rightleftharpoons Mg(s) + 2\,OH^-(aq)$	$OH^- \mid Mg(OH)_2 \mid Mg$	−2.687

[a] The difference between the $E°$ values defined with respect to 1 atm and 1 bar are so small that the values can usually be used interchangeably.